Experiment: 4

Data Modelling and Form Design for a Patient Appointment Booking Application in Zoho Creator

17th August 2026 | **Akshat Singh**
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AIM

To design and build a functional 'Patient Appointment Booking Form' application in Zoho Creator that demonstrates the correct usage of all thirteen Basic Fields available in the Form Builder — Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, Drop Down, Radio, Multi Select, and Checkbox — and to model the data structure of a real healthcare front-desk workflow, so that a patient can submit an appointment request online capturing personal identity, contact details, department preference, symptoms, existing conditions, preferred schedule, and consent for treatment.

PROBLEM STATEMENT

The front desk of a multi-specialty clinic receives appointment requests through phone calls and walk-ins, leading to double-bookings, missed follow-ups, and no structured record of a patient's symptoms or medical history before the consultation. Doctors often have no visibility into a patient's condition until the patient arrives. A low-code application built in Zoho Creator solves this by letting a patient submit a single online form that captures every detail the front desk and the doctor need in advance, removing the dependence on verbal phone intake.

PROCEDURE

1. Application and Form Creation

- Logged into Zoho Creator and opened the application 'project', created to host the clinic's patient-facing forms.
- Inside the application, opened the Form Builder and created a form named 'patient appointment booking form' to capture the complete appointment intake in one submission.
- This automatically created a linked report, 'patient appointment booking form Report', which lists every submitted appointment request for the front desk to review.

2. Drag-and-Drop of Basic Fields

Each of the thirteen Basic Fields was dragged from the palette onto the canvas and renamed to match the clinic's intake requirement, as mapped below.

Basic Field	Form Field Name	Purpose
Name	Patient_Name	Captures the patient's full name using First/Last Name sub-fields
Email	Email_Address	Captures a validated email address for sending appointment confirmations
Address	Residential_Address	Captures the patient's full residential address with city, state and pincode
Phone	Contact_Number	Captures a validated mobile number (India +91) for the clinic to call and confirm
Single Line	Referred_By	Short free-text entry for a referring doctor or source of referral
Multi Line	Symptoms_Description	Free-text description of the patient's current symptoms / reason for visit
Number	Age	Numeric entry with range validation for the patient's age
Date	Preferred_Appointment_Date	Date picker for the preferred consultation date
Time	Preferred_Appointment_Time	Time picker for the preferred consultation slot

241401008

Basic Field	Form Field Name	Purpose
Drop Down	Department	Single-select list of hospital departments / specializations
Radio	Gender	Single mandatory selection among mutually exclusive gender options
Multi Select	Existing_Conditions	Selection of one or more pre-existing medical conditions or allergies
Checkbox	Consent_For_Treatment	Captures the patient's consent to proceed with examination and treatment

3. Configuring Field Properties

- Selected the Patient_name field, enabled the Last Name display option alongside the default First Name sub-field, and marked its Data Privacy setting as 'Contains personal data (PII)' since patient names are personally identifiable.
- Configured the Email_address field with the built-in email-format validation left active, so confirmation emails are sent to a valid address only.
- Configured the Contact_number field with the country code defaulted to India (+91), matching the clinic's local patient base.
- Set the Age (Number) field to accept only realistic values, and marked Patient_name, Contact_number, Age, Gender, Department, and Preferred_appointment_date as mandatory.
- Marked the Consent_for_treatment checkbox as mandatory, so a patient cannot submit the form without agreeing to examination and treatment — directly addressing the clinic's need for documented consent before consultation.

4. Live Form Preview

Clicked the Preview icon in the Form Builder toolbar to view the patient-facing form and verified that all thirteen Basic Fields render in the correct order and type — Patient_name (First Name / Last Name split), Email_address, Residential_address (with Address Line 1/2, City, State, Postal Code and Country), Contact_number (with +91 country code), Referred_by, Symptoms_description, Age, Preferred_appointment_date, Preferred_appointment_time, Department, Gender, Existing_conditions, and Consent_for_treatment.

5. Submitting a Sample Record and Reviewing the Report

- Filled and submitted a sample appointment request for patient Meera Krishnan, entering her contact details, a referring physician, and the symptom description 'Persistent headache and mild fever for the past 3 days.'
- Opened the 'patient appointment booking form Report' and confirmed the record appears with Patient_name, Email_address and Symptoms_description visible in the list view.
- Opened the record individually and verified that every field — Preferred_appointment_date, Preferred_appointment_time, Department, Gender, Consent_for_treatment, Residential_address, Contact_number, Referred_by, Age, and Existing_conditions — was correctly stored and retrievable, giving the front desk and the doctor full visibility into the patient's condition before the visit.

6. Data Model — Entity-Relationship Diagram

To formalize the underlying data structure captured by the form, the fields were modelled as two related entities — PATIENT and APPOINTMENT — connected through a 'Books' relationship, reflecting how a single patient can book one or more appointments over time. Patient identity and contact fields sit on the PATIENT entity, while scheduling, department, and clinical-intake fields sit on the APPOINTMENT entity.

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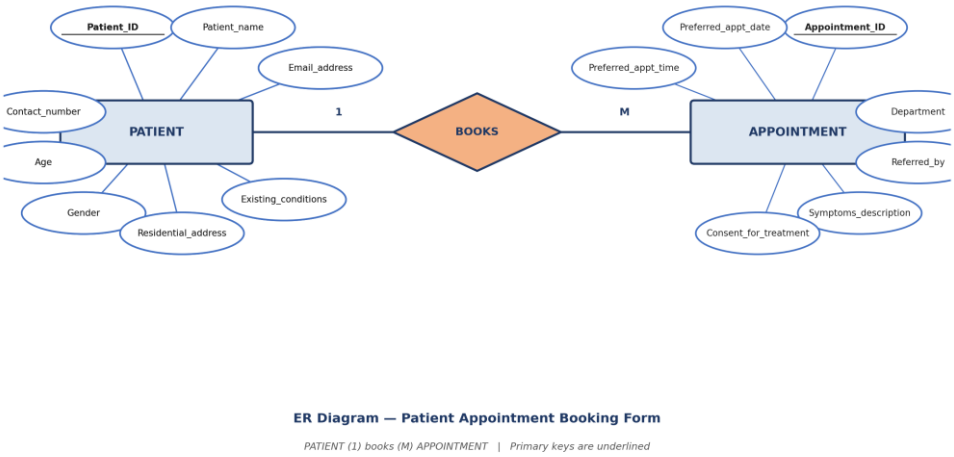


Fig 1. Entity-Relationship Diagram — PATIENT (1) books (M) APPOINTMENT

OUTPUT

The following screenshots capture the completed application: the Form Builder showing the complete field configuration, the live patient-facing form with a sample entry, and the generated report showing the stored appointment record.

1. Form Builder View — Field Configuration

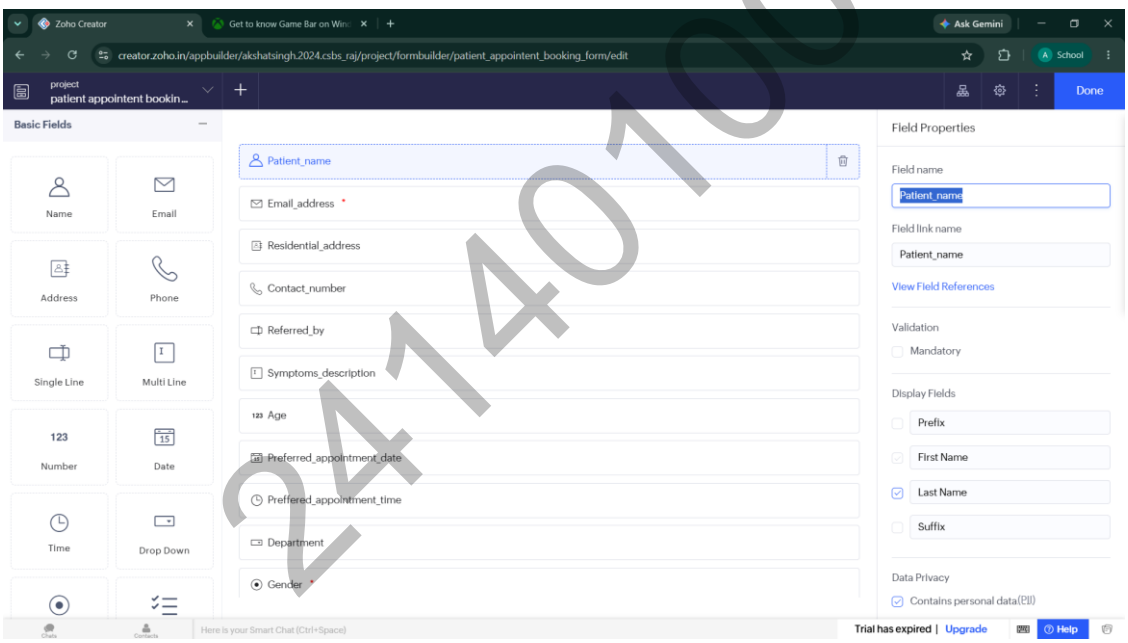


Fig 2. Form Builder — Basic Fields palette showing all thirteen fields of the patient appointment booking form

2. Live Form — Sample Patient Entry

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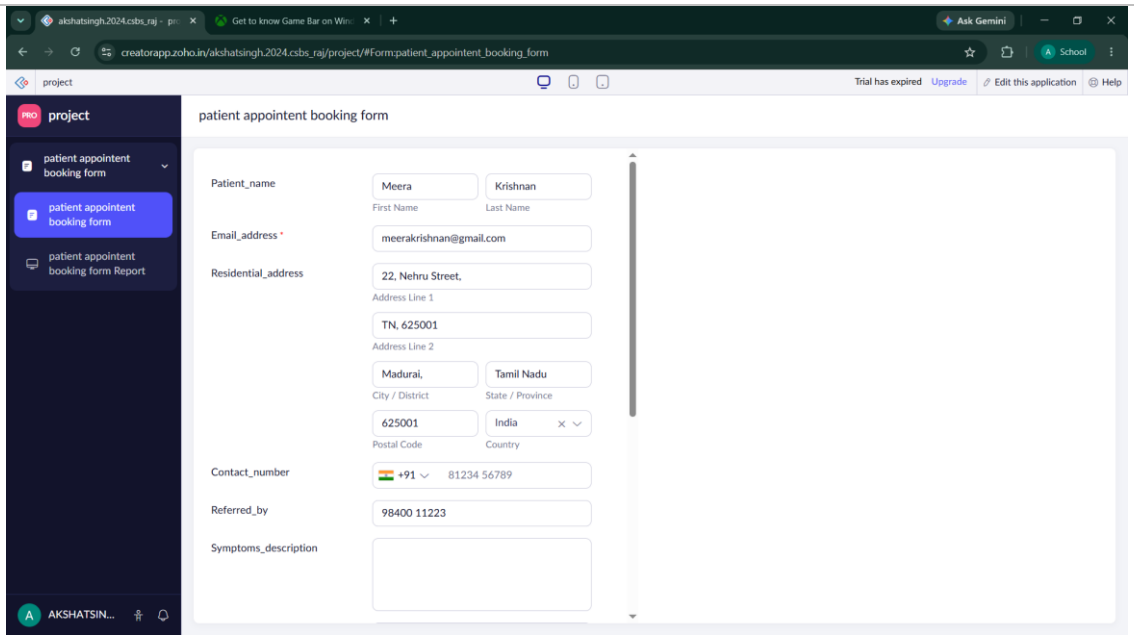


Fig 3. Live Preview — patient-facing form filled with a sample record (Meera Krishnan)

3. Report View — Submitted Appointment Record

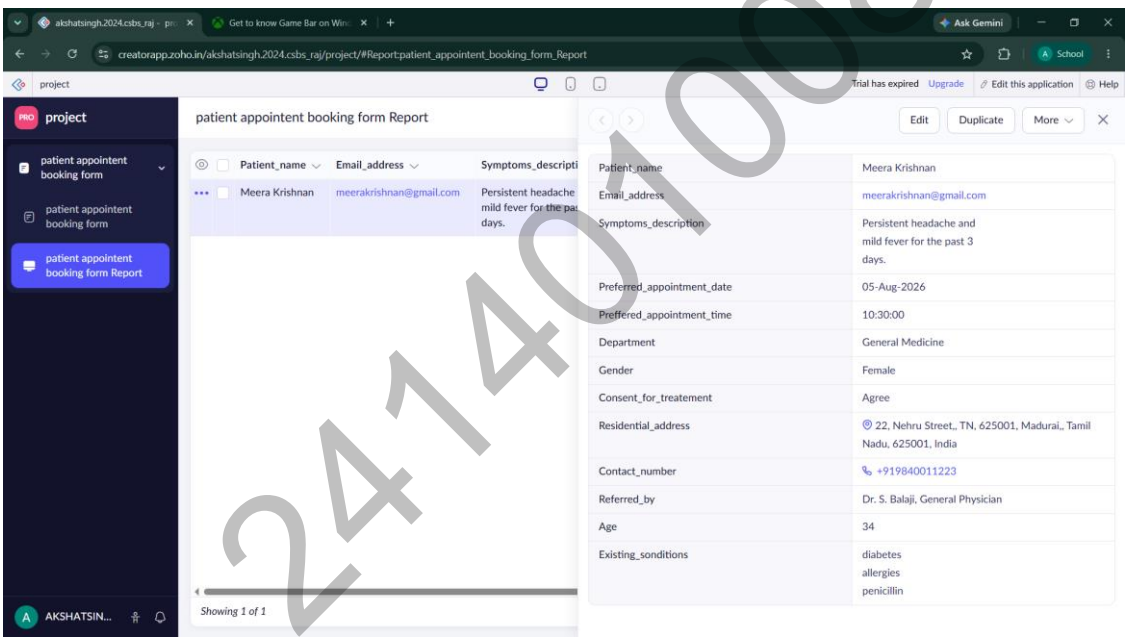


Fig 4. Appointment Report — detail view of the stored record, visible to the front desk and doctor before consultation

RESULT

A fully functional Patient Appointment Booking application was successfully designed and built in Zoho Creator using all thirteen Basic Fields. The form correctly captures patient identity, contact details, department preference, symptoms, existing conditions, preferred schedule, and consent for treatment in a single online submission, and the auto-generated report accurately reflects every submitted record. This eliminates the clinic's reliance on unstructured phone and walk-in intake, prevents double-bookings, and gives doctors visibility into a patient's condition ahead of the consultation — confirming that the low-code solution meets the stated problem statement.

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